

Roll No.

Y – 3960

**MCA (First Semester) EXAMINATION,
Dec.-2021**

Paper-MCA-104

**COMPUTER ORGANIZATION AND
ASSEMBLY LANGUAGE
PROGRAMMING**

Time : Three Hours

Maximum Marks : 60

Minimum Pass Marks : 21

Note– Attempt *all* questions. Question No. 1 is compulsory.

1. Attempt any *five* questions (Not more than 50 words each)---

➤ (i) What do you mean by number system ?

2

(ii) Explain addressing modes.

2

(iii) What do you understand by Time Shared Common bus ?

2

(iv) What do you mean by character code ?

2

(v) Explain memory stack.

2

P.T.O.

- (vi) What do you mean by Adder ? 2
- (vii) What do you understand by control word ? 2
- (viii) What is Strobe control ? 2
- (ix) What is Instruction Pipeline ? 2
- (x) What do you understand by static RAM ? 2

Unit-I

2. Discuss Basic Computer Organization. Give details of major components like bus, memory, CPU, control and logic unit. 10

Or

How the computer instructions are timed and controlled ?

Unit-II

3. Explain all the CPU registers with their size and functions. 10

Or

- What is the Purpose of seven-segment decoder circuit ? What is a “seven-segment” display, and why do we need a decoder circuit to drive it.

Unit-III

4. List two methods of data transfer. 10

(3)

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Or

- Explain the various arithmetic and logical operations alongwith register transfer.

Unit-IV

5. What do you understand by Processor's instruction set ? Also explain addressing modes for 8086 instructions. 10

Or

- What is interrupt ? Explain the various types of interrupt with suitable example.

Unit-V

6. Explain the working of associative memory. Describe block representation of an associative memory with figure. 10

Or

Write an assembly language Program for calculating the factorial of a number using 8086 microprocessor.